

Multiple Choice

1. **Answer:** B

Options: A) `replace(old, new [, max])`; B) `strip([chars])`; C) `swapcase()`; D) `title()`

Note: The ‘`strip([chars])`’ function in Python 3 is designed specifically to remove all leading and trailing whitespace from a string, making it the correct answer for this question.

2. **Answer:** A

Options: A) 1; B) 2; C) 3; D) 4

Note: The correct answer is 1 because the ‘`min()`’ function in Python 3 returns the smallest element in a list, and in the list `l = [1, 2, 3, 4]`, the smallest value is 1.

3. **Answer:** A

Options: A) $O(1)$; B) $O(n)$; C) $O(n^2)$; D) $O(\log n)$

Note: $O(1)$ is considered the smallest asymptotic notation because it represents a constant time complexity that does not grow with the size of the input, making it the most efficient in terms of performance.

4. **Answer:** B

Options: A) A device driver is assigned to the device.; B) An Internet Protocol (IP) address is assigned to the device.; C) A packet number is assigned to the device.; D) A Web site is assigned to the device.

Note: When a new device connects to the Internet, it is assigned an Internet Protocol (IP) address, which uniquely identifies it on the network and enables communication with other devices.

5. **Answer:** C

Options: A) $O(1)$; B) $O(n)$; C) $O(n^2)$; D) $O(\log n)$

Note: $O(n^2)$ is the largest asymptotic notation among the options provided because it grows faster than linear ($O(n)$) or logarithmic ($O(\log n)$) functions as the input size n increases, indicating that it represents a higher rate of growth in computational complexity.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The sum of the list `l = [1, 2, 3, 4]` in Python 3 is 10 because adding the numbers together ($1 + 2 + 3 + 4$) equals 10.

7. **Answer:** True

Options: A) True; B) False

Note: In Python 3, the expression `x << 3` performs a left bitwise shift on the binary representation of `x`, which multiplies it by 2 raised to the power of 3; therefore, when `x` is 1 (binary 0001), shifting it left by 3 results in 8 (binary 1000).

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the expression `['Hi!'] * 4` successfully creates a list containing four references to the string "Hi!" without raising any syntax errors.

9. **Answer:** False

Options: A) True; B) False

Note: Averaging numeric values may mask outliers and inaccuracies, making it an ineffective method for detecting data entry errors, as it does not identify individual discrepancies in the dataset.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because $O(n)$ represents a linear growth rate, which increases more slowly than $O(n^2)$, a quadratic growth rate, as the input size increases.

Long Form Response

11. **Answer:** `def series_sum(number): | return total`

Note: The answer correctly identifies the need for a function that calculates the sum of squares of the first `n` integers, although it lacks the implementation details to compute the total.

12. **Answer:** `def chinese_zodiac(year): | return sign`

Note: correct because it outlines the structure of a function named 'chinese_zodiac' that takes a year as input and is intended to return the corresponding zodiac sign, capturing the essential requirement of the question.

End of Answer Key